

IS 3001

GROUP PROJECT REPORT

GROUP 16



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1.0 INTRODUCTION

The dataset used in this project named as "Insurance Claim Analysis: Demographic and Health". It contains valuable information about 'Insurance Claims', offering insights into the patterns among those who file these claims. By looking at key factors like age, gender, BMI, blood pressure, diabetic status, number of children, smoking status, and region, we can understand how different demographics and health traits impact insurance claims. The dataset was obtained from the 'Kaggle' website and credit goes to 'Sumit Kumar Shukla'.

The dataset includes details for each person, identified by a unique ID. It records the person's age, categorizes them into one of four regions (Southeast, Southwest, Northeast, Northwest), and notes the number of children they have. It also captures information on whether the person has diabetes. By analyzing these factors, we can explore how age, gender, BMI, blood pressure, and other health characteristics relate to the likelihood of making an insurance claim.

This project uses three different sampling methods named as Simple Random Sampling, Stratified Sampling, and Two-Stage Cluster Sampling. In this project we used R software for analyzing.

We will compare the estimates obtained from these sampling methods to the actual values in the dataset. Additionally, we conduct a graphical analysis to highlight the findings and trends in the data.

The following sections will explain each sampling method, the steps taken for analysis, and the results.

Link for the data set:

<https://www.kaggle.com/datasets/thedevastator/insurance-claim-analysis-demographic-and-health>

2.0 METHODOLOGY

I. Sample Size Calculation

1. Simple Random Sampling (SRS)

Simple Random Sampling is a basic form of probability sampling technique in which every possible subset of n distinct units in the population has the same probability of being selected as the sample. Each sample is equally likely. Following equations are used to calculate the sample size.

$$n = \frac{no}{1 + \left(\frac{no}{N}\right)} \quad \text{where} \quad no = \left(Z \alpha/2 \frac{s}{e}\right)^2$$

n - sample size

N - population size

S -population standard deviation

e -margin of error

$Z \alpha/2$ - Z value of the confidence level $(1 - \alpha)$

In our project we use R soft wear to calculate sample size. We used the ‘rsampcalc’ function included in a package named as ‘sampler’ to calculate the sample size. We plan to take an SRS from 1332 observations using a 95% CI with margin of error 0.03. this is the code we used,

```
n <- rsampcalc(nrow(data), e = 3, ci = 95)
```

using this code, we estimate that the sample size is 593. To draw simple random samples of we use this code,

```
rsamp(df=data, n = n, rep = FALSE)
```

2. Stratified Sampling

In stratified sample technique the population is partitioned into strata. Observations are selected from each stratum proportionally to the size of each stratum to obtain the sample size of n. Simple random sample size was calculated using 'rsampcalc' function in 'sampler' package. the value we got for sample size is 593. Then by using simple random sampling method, samples were selected from each stratum according to weights to get the total sample size of 593. For that 'ssampcalc' function in 'sampler' package is used.

This is the code we used,

```
ssampcalc (df=data, n=n, strata =smoker)
```

	smoker	Nh	wt[,1]	nh[,1]
	<chr>	<int>	<dbl>	<dbl>
1	No	1058	0.794	471
2	Yes	274	0.206	122

we take 471 observations from nonsmokers and 122 observations from smokers for sample. To draw samples we use this code,

```
ssamp(df=data,n=n,strata=smoker,over=1)
```

3. Two-stage Cluster Sampling

In cluster sampling, Population is divided into clusters. First, we choose Some clusters randomly. we use the inbuilt function ‘sample’ to obtain random clusters. Here we obtained two clusters from four clusters. We used ‘dplyr’ package in R. we filter out the randomly selected two clusters from our dataset. From this filtered sample, we obtain a random sample that contains individuals from each randomly selected cluster. Total sample size calculated using rsampcalc function. In two stage cluster sampling, sample size of each cluster is calculated according to proportion of each cluster in population.

	region <chr>	cluster_count <int>	sample_size <dbl>
1	northeast	231	204
2	southeast	442	389

II. Stratification and Clustering variables

1. Stratification variables

In stratification, target population should be partitioned into strata. the stratification should be done on variables which has a significantly impact the outcome variable. When selecting the stratification variable, it also considered that strata do not overlap and the variance within the strata is minimum. In our dataset smoking status has a significantly impact to the insurance claim amount. nonsmokers claim amount is higher than smokers claim amount. Smoking status is not overlap.

Variance within the stratum (nonsmokers and smokers) is small, so in this case we selected smoking status as the stratified variable. Smoking status is a categorical variable and it has two strata as nonsmokers and smokers. In population there are 1058 nonsmokers and 274 smokers. We draw a sample of 471 nonsmokers and 122 smokers.

2.Cluster variables

Cluster sampling is a probability sampling technique where the entire population is divided into distinct groups, known as clusters. These clusters are typically based on geographical regions or organizations and should ideally be similar to each other in terms of their characteristics. Cluster sampling is useful when a population is too large and widespread, making it impractical to conduct simple random sampling.

Cluster Variables are used to group individuals into clusters that share common characteristics. In this project, we have chosen the “region” as our clustering variable, which divides the data into four clusters as northeast, northwest, southeast, and southwest. There is no more variability between these regions.

northeast	northwest	southeast	southwest
231	345	442	314

In two-stage cluster sampling, the process involves two key steps. First, randomly select a subset of clusters from the total number of clusters. In our case, we randomly chose the northeast and northwest clusters. Then within the selected clusters, we randomly sample individuals or units. This approach helps reduce costs.

3.0 RESULTS OF THE STUDY

1.Simple Random Sampling

	Sample1		Sample2		Population	
	Estimated population mean	SE	Estimated population mean	SE	population mean	SE
Age	36.921	0.4634	37.59	0.4605	38.08634	0.304489
BMI	30.659	0.2434	30.581	0.2464	30.65833	0.1676587
Blood pressure	93.906	0.4724	93.899	0.4636	94.18919	0.3135958
Num. of children	1.1029	0.0492	1.1484	0.0502	1.09985	0.03304305
Claim amount	12804	498.14	12945	487.91	13325.25	331.8016
	Estimated population Proportion	SE	Estimated population Proportion	SE	population Proportion	SE
Gender						
Female	0.4688	0.0205	0.47723	0.0205	0.496997	
Male	0.5312	0.0205	0.52277	0.0205	0.503003	
Diabetic						
No	0.51602	0.0205	0.51433	0.0205	0.5217718	
Yes	0.48398	0.0205	0.48567	0.0205	0.4782282	
Smoker						
No	0.80776	0.0162	0.8027	0.0164	0.7942943	
Yes	0.19224	0.0162	0.1973	0.0164	0.2057057	

Region						
Northeast	0.17032	0.0154	0.18044	0.0158	0.1734234	
Northwest	0.25632	0.0179	0.26813	0.0182	0.2590090	
Southeast	0.32715	0.0193	0.30860	0.0190	0.3318318	
Southwest	0.24621	0.0177	0.24283	0.0176	0.2357357	
	Estimated population Total	SE	Estimated population Total	SE	population Total	
Gender						
Female	624.45	27.319	635.68	27.344	662	
Male	707.55	27.319	696.32	27.344	670	
Diabetic						
No	687.34	27.358	685.09	27.361	695	
Yes	644.66	27.358	646.91	27.361	637	
Smoker						
No	1075.93	21.573	1069.19	21.786	1058	
Yes	256.07	21.573	262.81	21.786	274	
Region						
Northeast	226.87	20.579	240.34	21.052	231	
Northwest	341.42	23.902	357.15	24.251	345	
Southeast	435.76	25.685	411.06	25.288	442	
Southwest	327.95	23.584	323.45	23.474	314	
Regression						
Claim amount	12959.83		13104.23		13325.25	

- According to above findings, In the simple random sampling, we can see the Estimated mean of age, BMI, Blood pressure, Number of children are approximately equal to the population parameters of those variables. But insurance Claim amount is little bit different from the population values.
- When considering the categorical variables, estimated proportions of categories are approximately equal from the population values
- Estimated totals of categories are little bit different from the population values in some variables.
- In regression estimation there is little bit difference between estimated values and population values.

2.Stratified Sampling

	Sample1		Sample2		population	
	Estimated population mean	SE	Estimated population mean	SE	Mean	SE
Age	34.826	0.5458	35.045	0.5476	38.08634	0.304489
BMI	30.996	0.2705	31.201	0.2546	30.65833	0.1676587
Blood pressure	94.241	0.4715	93.828	0.4024	94.18919	0.3135958
Children	1.208	0.0511	1.1529	0.0424	1.09985	0.03304305
Claim amount	13556	318.87	13239	295.88	13325.25	331.8016

	Estimated population Proportion	SE	Estimated population Proportion	SE	population Proportion	SE
Gender						
Female	0.38463	0.02	0.37604	0.0198	0.496997	
Male	0.61537	0.02	0.62391	0.0198	0.503003	
Diabetic						
No	0.51715	0.0205	0.53196	0.0205	0.5217718	
Yes	0.48285	0.0205	0.46804	0.0205	0.4782282	
Smoker						
No	0.79429	0	0.79429	0	0.7942943	
Yes	0.20571	0	0.20571	0	0.2057057	
Region						
Northeast	0.20366	0.0165	0.1934205	0.0162	0.1734234	
Northwest	0.26926	0.0181	0.2442283	0.0177	0.2590090	
Southeast	0.34476	0.0195	0.3315758	0.0193	0.3318318	
Southwest	0.18232	0.0158	0.2290818	0.0173	0.2357357	
	Estimated population Total	SE	Estimated population Total	SE	population Total	
Gender						
Female	512.33	26.599	500.95	26.345	662	
Male	819.67	26.599	831.05	26.345	670	
Diabetic						
No	688.84	27.37	708.57	27.317	695	
Yes	643.16	27.37	623.43	27.317	637	

Smoker						
No	1058	0	1058	0	1058	
Yes	274	0	274	0	274	
Region						
Northeast	271.28	21.971	257.6361	21.6014	231	
Northwest	358.66	24.064	325.3120	23.5174	345	
Southeast	459.22	25.973	441.6590	25.7704	442	
Southwest	242.85	21.001	305.1370	23.0031	314	
Regression						
Claim Amount	13297.06		13529.66		13325.25	

- In the stratified sampling, we can see the Estimated mean of age, BMI, Blood pressure, Number of children are approximately equal to the population parameters of those variables. Insurance Claim amount is little bit different from the population values. But stranded error of it is less than the simple random sampling.
- When considering the categorical variables, estimated proportions of Gender category is little bit different from the population values. But in other categories estimated values and population values are approximately equal.
- Estimated totals of categories are little bit different from the population values in some variables.
- In regression estimation there is little difference between estimated values and population values.

3.Two-stage Cluster Sampling

	Sample1		Sample2		population	
	Estimated population mean	SE	Estimated population mean	SE	Mean	SE
Age	38.274	0.5875	37.701	0.1297	38.08634	0.304489
BMI	30.66	1.5909	30.007	0.832	30.65833	0.1676587
Blood pressure	95.13	0.958	93.43	0.5592	94.18919	0.3135958
Children	1.0586	0.039	1.1492	0.0557	1.09985	0.03304305
Claim Amount	14779	2067.2	12176	677.88	13325.25	331.8016
	Estimated population Proportion	SE	Estimated population Proportion	SE	population Proportion	SE
gender						
female	0.49389	0.0331	0.49328	0.0191	0.496997	
male	0.50611	0.0331	0.50672	0.0191	0.503003	
diabetic						
No	0.49224	0.0168	0.53744	0.0109	0.5217718	
Yes	0.50776	0.0168	0.46256	0.0109	0.4782282	
Smoker						
No	0.75525	0.0494	0.82395	0.0148	0.7942943	
Yes	0.24475	0.0494	0.17605	0.0148	0.2057057	

	Estimated population Total	SE	Estimated population Total	SE	population Total	
gender						
female	664.77	44.558	650.14	25.157	662	
male	681.23	44.558	667.86	25.157	670	
diabetic						
No	662.56	22.55	708.35	14.422	695	
Yes	683.44	22.55	609.65	14.422	637	
Smoker						
No	1016.57	66.455	1085.96	19.455	1058	
Yes	329.43	66.455	232.04	19.455	274	
Regression						
Claim Amount	14775.59		14703.75		13325.25	

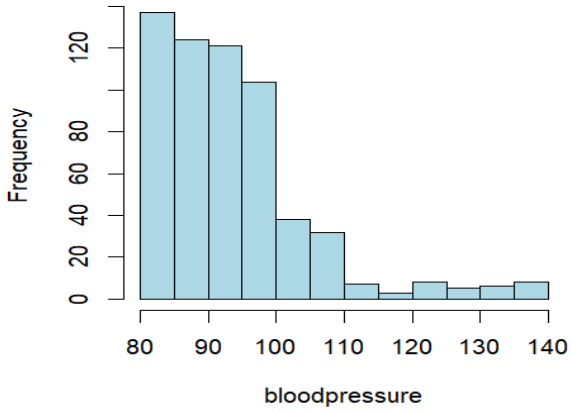
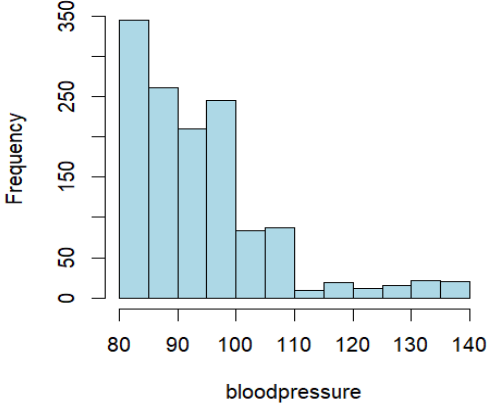
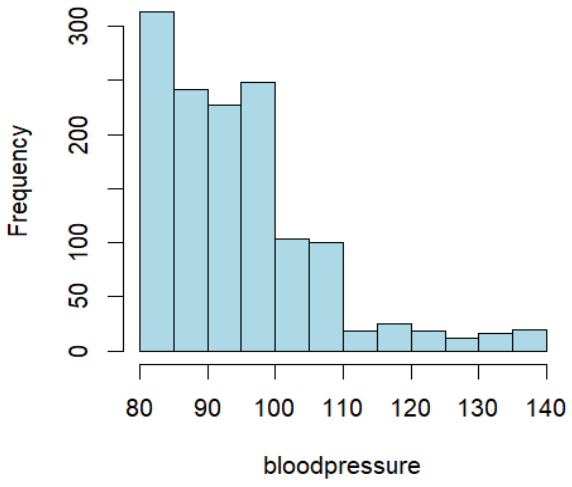
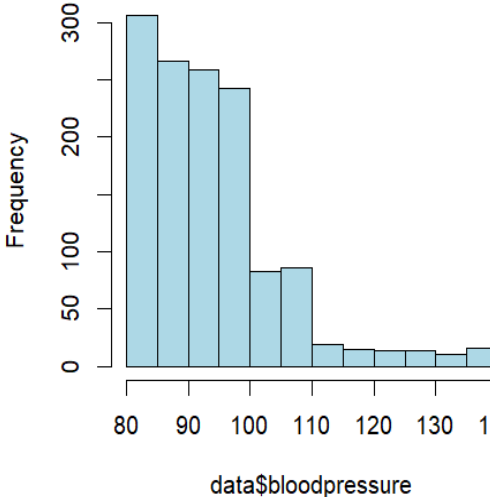
- In the Two-stage Cluster Sampling, we can see the Estimated mean of age, BMI, Blood pressure and Number of children are approximately equal to the population parameters of those variables. But insurance Claim amount is little different from the population values.
- When considering the categorical variables, estimated proportions of categories and population proportion values are approximately equal.
- Estimated totals of categories are little bit different from the population values in some variables.
- In regression estimation there is little difference between estimated values and population values.

4.0 GRAPHICAL ANALYSIS

Histogram for Insurance Claim Amount

Simple random sampling	Stratified sampling
Histogram of claim	Histogram of claim
Cluster sampling	Population
Histogram of claim	Histogram of data\$claim
We observe that the majority of individuals have claim amounts of less than \$15,000, while only a few individuals have claim amounts exceeding \$50,000.	

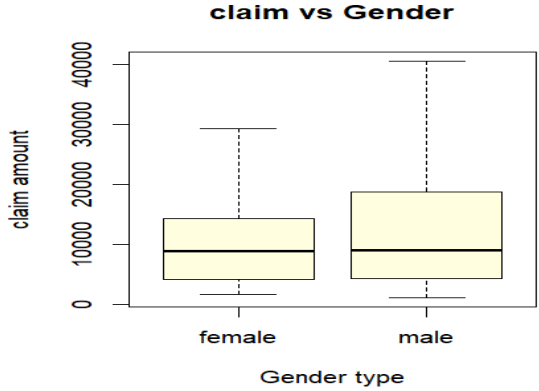
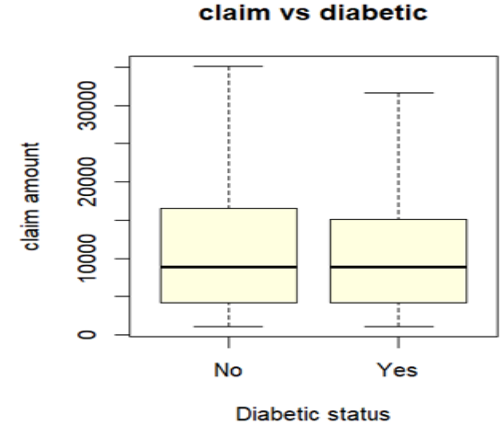
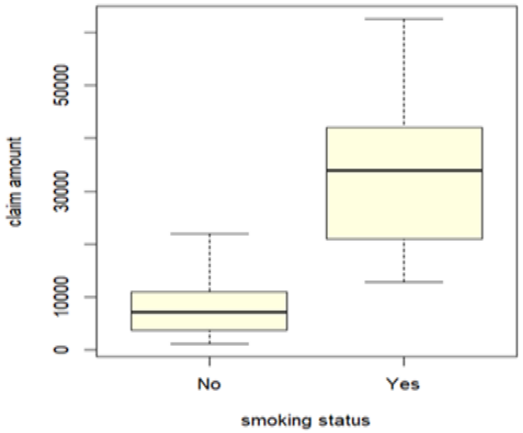
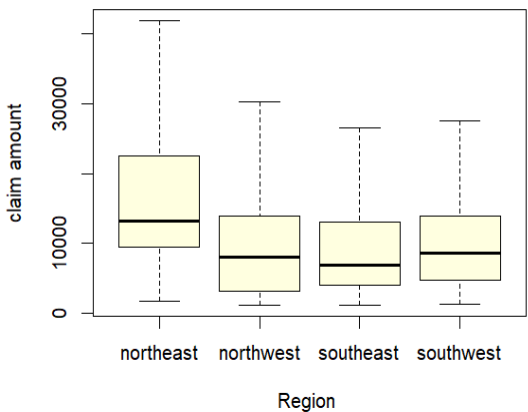
Histogram for Insurance Blood pressure

Simple random sampling	Stratified sampling
Histogram of bloodpressure 	Histogram of bloodpressure 
Cluster sampling	population
Histogram of bloodpressure 	Histogram of data\$bloodpressure 
We can see that the majority of individuals have bloodpressure less than 100, while only a few individuals have bloodpressure higher than 110.	

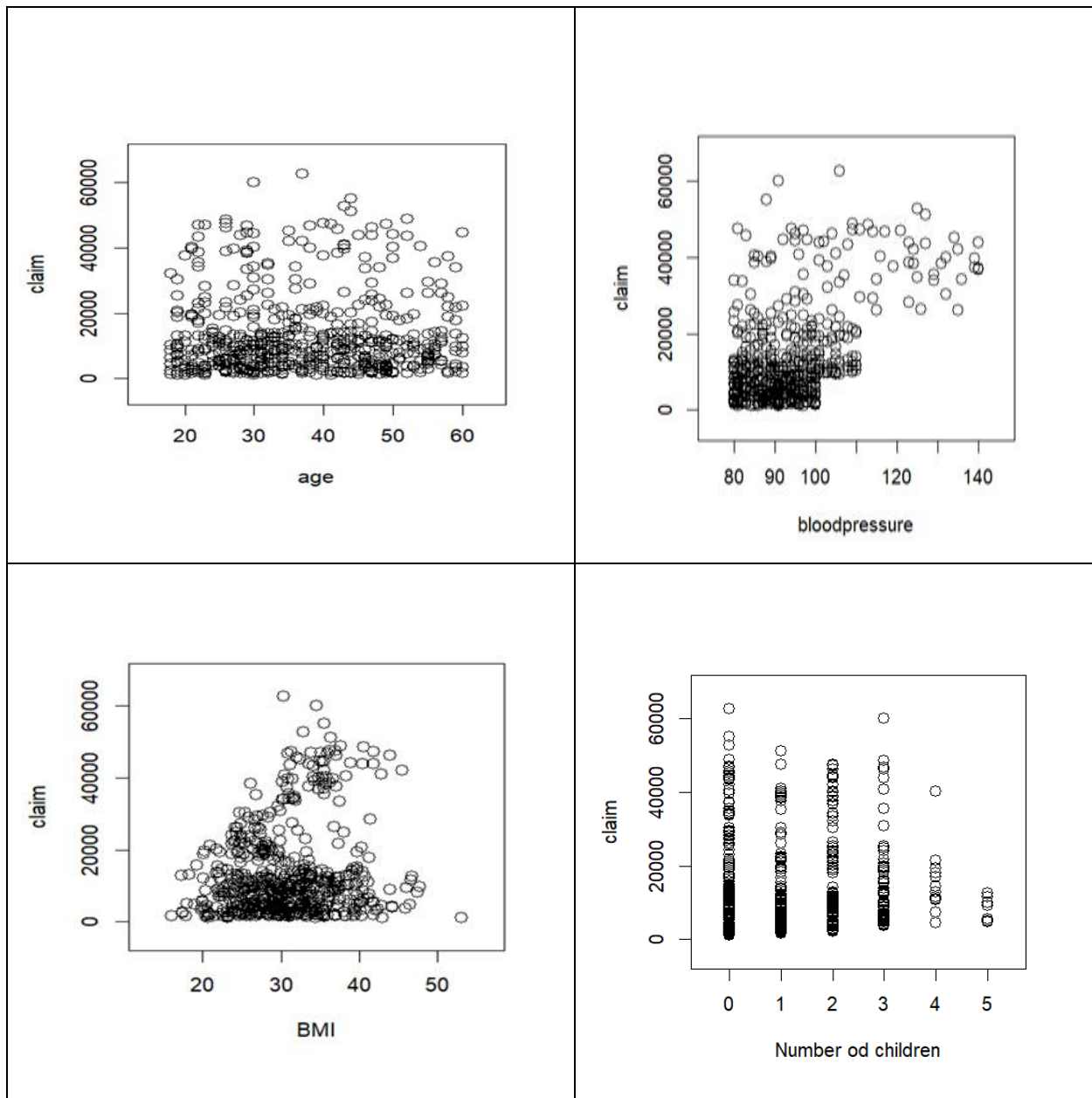
Histogram for Insurance BMI

Simple random sampling	Stratified sampling
Histogram of bmi	Histogram of bmi
Cluster sampling	population
Histogram of bmi	Histogram of data\$bmi
We observe that the majority of individuals have bmi between 20-40. only a few individuals have bmi less than 20 and more than 40.	

Boxplots for categorical variables

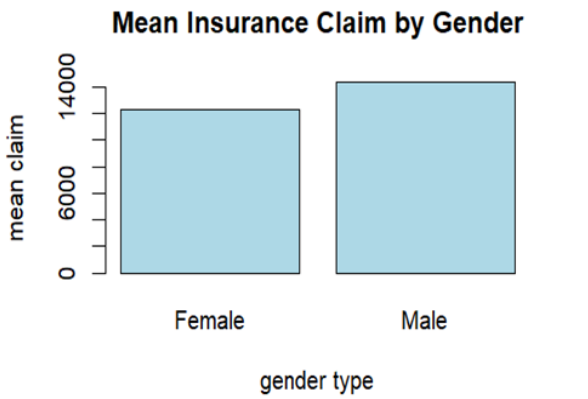
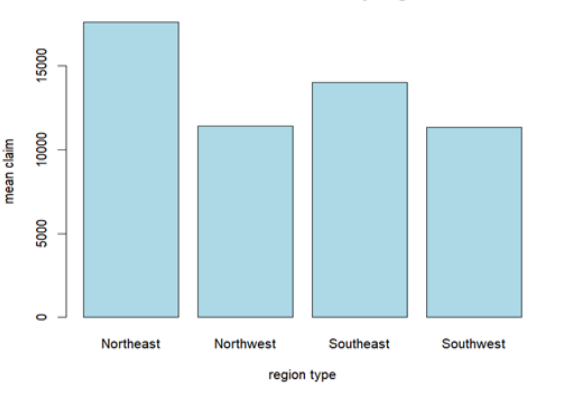
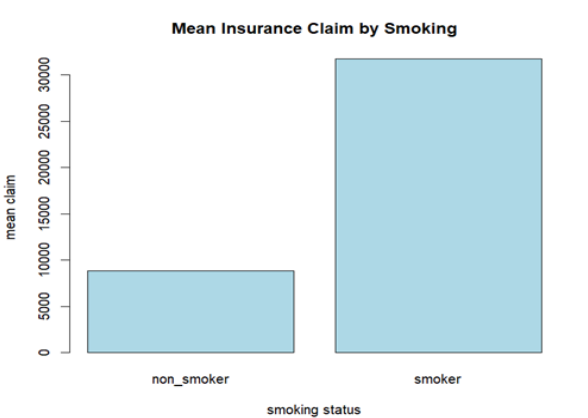
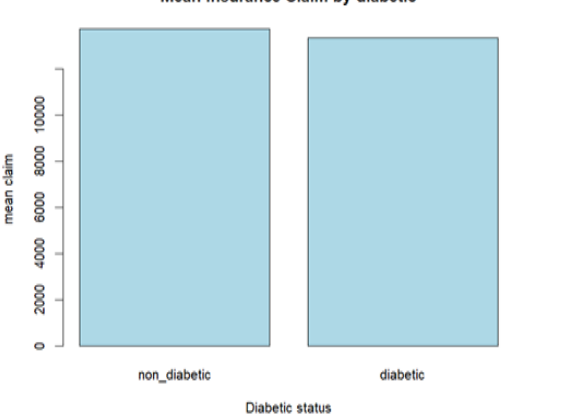
 claim vs Gender claim amount female male Gender type	 claim vs diabetic claim amount No Yes Diabetic status
We can see mean claim amount of males and females are approximately equal. The maximum claim amount of male is higher than females.	Mean claim amount of diabetic and nondiabetic are approximately equal. The maximum claim amount of nondiabetic is higher than diabetic.
 claim vs smoking claim amount No Yes smoking status	 claim vs region claim amount northeast northwest southeast southwest Region
We can see mean claim amount and maximum claim amount of smokers are higher than the non smokers.	Mean claim amount and maximum claim amount of northeast is higher than other regions

Scatter plots



The plot of bloodpressure vs claim show that there is a roughly positive linear relationship between them. The plots of claim vs age, bmi and number of children show that there's no special linear relationship between them.

Bar charts for categorical variables

 Mean Insurance Claim by Gender <table border="1"><thead><tr><th>gender type</th><th>mean claim</th></tr></thead><tbody><tr><td>Female</td><td>~12000</td></tr><tr><td>Male</td><td>~12000</td></tr></tbody></table>	gender type	mean claim	Female	~12000	Male	~12000	 Mean Insurance Claim by Region <table border="1"><thead><tr><th>region type</th><th>mean claim</th></tr></thead><tbody><tr><td>Northeast</td><td>~18000</td></tr><tr><td>Northwest</td><td>~11500</td></tr><tr><td>Southeast</td><td>~14000</td></tr><tr><td>Southwest</td><td>~11500</td></tr></tbody></table>	region type	mean claim	Northeast	~18000	Northwest	~11500	Southeast	~14000	Southwest	~11500
gender type	mean claim																
Female	~12000																
Male	~12000																
region type	mean claim																
Northeast	~18000																
Northwest	~11500																
Southeast	~14000																
Southwest	~11500																
We can see mean claim amount of males and females are approximately equal	Mean claim amount of northeast is much higher than other regions																
 Mean Insurance Claim by Smoking <table border="1"><thead><tr><th>smoking status</th><th>mean claim</th></tr></thead><tbody><tr><td>non_smoker</td><td>~8000</td></tr><tr><td>smoker</td><td>~28000</td></tr></tbody></table>	smoking status	mean claim	non_smoker	~8000	smoker	~28000	 Mean Insurance Claim by diabetic <table border="1"><thead><tr><th>Diabetic status</th><th>mean claim</th></tr></thead><tbody><tr><td>non_diabetic</td><td>~11000</td></tr><tr><td>diabetic</td><td>~11000</td></tr></tbody></table>	Diabetic status	mean claim	non_diabetic	~11000	diabetic	~11000				
smoking status	mean claim																
non_smoker	~8000																
smoker	~28000																
Diabetic status	mean claim																
non_diabetic	~11000																
diabetic	~11000																
We can see mean claim amount of smokers are higher than the non smokers.	We can see mean claim amount of diabetic and non diabetic are approximately equal.																

5.0 CONCLUSION OF THE ANALYSIS

- The insurance claim dataset has provided valuable insights into how various demographic and health-related factors influence claim amounts. By using different sampling designs (Simple random sampling, stratified sampling, and two-stage cluster sampling), we were able to estimate key statistics such as the mean, proportion and total of the variables along with their standard errors.
- In simple random sampling, our findings indicate that the estimated mean values and proportions of samples 1 and sample 2 are approximately equal. Also, the regression estimate values of the two samples are much similar.

When we compare the estimates mean values of samples and population, we can see they are approximately equal but insurance Claim amount is little bit different from the population values.

- In stratified sampling, Estimated mean values and proportions of samples 1 and sample 2 are approximately equal. Also, the regression estimate values of the two samples are much equal.

The estimates mean values of samples and population are approximately equal than the simple random sampling. Also, the proportions are much equal. In regression estimation there is no much difference between estimated values and population values

- In Two stage cluster sampling, our findings indicate that the estimated mean values and proportions of samples 1 and sample 2 are approximately equal. Also, the regression estimate values of the two samples are much similar.

When we compare the estimates mean values of samples and population, we can see they are approximately equal but insurance Claim amount is little different from the population values. Regression estimated value for the claim amount is also little deferent form population value.

- When comparing the estimated values, we can conclude that stratified sampling provides the most accurate estimates compared to the other sampling methods.

6.0 R CODE

#Import the data set

```
insurance_data = read.csv("c:\\Insurance_data.csv",header = TRUE)
head(insurance_data)
```

#clean the data set

```
data <- insurance_data[complete.cases(insurance_data) & !apply(insurance_data
== "", 1, any), ]
nrow(data)
summary(data)
```

#Install "survey" package

```
install.packages("survey")
library(survey)
```

###-----Simple Random Sample-----###

#Calculate Sample size

```
install.packages("sampler")
library(sampler)
n<- rsampcalc(nrow(data), e=3, ci=95)
n
```

#draw the sample1

```
set.seed(12345)
SRS_Sample1 = rsamp(df = data, n=n, rep = FALSE)
SRS_Sample1
```

```
#-----Sample1-----#
```

```
#obtain weights
```

```
SRS_Sample1$weight <- nrow(data) / n
```

```
#defining the survey design object
```

```
SRS_Sample1_design= svydesign(id=~1, weights=SRS_Sample1$weight , data=
SRS_Sample1)
```

```
summary(SRS_Sample1_design)
```

```
#Estimating the mean values
```

```
svymean(~age,SRS_Sample1_design)
```

```
svymean(~bmi,SRS_Sample1_design)
```

```
svymean(~bloodpressure,SRS_Sample1_design)
```

```
svymean(~children,SRS_Sample1_design)
```

```
svymean(~claim,SRS_Sample1_design)
```

```
#Estimating the proportions
```

```
svymean(~gender,SRS_Sample1_design)
```

```
svymean(~diabetic,SRS_Sample1_design)
```

```
svymean(~smoker,SRS_Sample1_design)
```

```
svymean(~region,SRS_Sample1_design)
```

```
#Estimating the totals
```

```
svytotal(~gender,SRS_Sample1_design)
```

```
svytotal(~diabetic,SRS_Sample1_design)
```

```
svytotal(~smoker,SRS_Sample1_design)
```

```
svytotal(~region,SRS_Sample1_design)
```

#regression estimations

```
plot(SRS_Sample1$claim,SRS_Sample1$age)
plot(SRS_Sample1$claim,SRS_Sample1$bmi)
plot(SRS_Sample1$claim,SRS_Sample1$bloodpressure)
plot(SRS_Sample1$claim,SRS_Sample1$children)
SLR1 = lm(claim ~ bloodpressure, data = SRS_Sample1)
summary(SLR1)
mean_Claim = coef(SLR1)[1] + coef(SLR1)[2]*mean(data$bloodpressure)
mean_Claim
```

#-----Sample2-----#

#draw the sample2

```
set.seed(23456)
SRS_Sample2 = rsamp(df = data, n=n, rep = FALSE)
SRS_Sample2#Import the data set
insurance_data = read.csv("c:\\Insurance_data.csv",header = TRUE)
head(insurance_data)
```

#obtain weights

```
SRS_Sample2$weight <- nrow(data) / n
```

#defining the survey design object

```
SRS_Sample2_design= svydesign(id=~1, weights=SRS_Sample2$weight , data=
SRS_Sample2)
summary(SRS_Sample2_design)
```

#Estimating the mean values

```
svymean(~age,SRS_Sample2_design)
svymean(~bmi,SRS_Sample2_design)
svymean(~bloodpressure,SRS_Sample2_design)
svymean(~children,SRS_Sample2_design)
svymean(~claim,SRS_Sample2_design)
```

#Estimating the proportions

```
svymean(~gender,SRS_Sample2_design)
svymean(~diabetic,SRS_Sample2_design)
svymean(~smoker,SRS_Sample2_design)
svymean(~region,SRS_Sample2_design)
```

#Estimating the totals

```
svytotal(~gender,SRS_Sample2_design)
svytotal(~diabetic,SRS_Sample2_design)
svytotal(~smoker,SRS_Sample2_design)
svytotal(~region,SRS_Sample2_design)
```

#regression estimations

```
SLR2 = lm(claim ~ bloodpressure, data = SRS_Sample2)
summary(SLR2)
mean_Claim = coef(SLR2)[1] + coef(SLR2)[2]*mean(data$bloodpressure)
mean_Claim
```

###-----Stratified Sampling-----###

#Calculate Sample size

```
n_strata<-ssampcalc(df=data, n=n, strata =smoker)
```

```
n_strata
```

###-----sample1-----###

#draw the sample1

```
set.seed(12346)
```

```
strat_sample1=ssamp(df=data,n=n,strata=smoker,over=1)
```

```
strat_sample1
```

#obtaining weight

```
strat_data1$weight <- table(data$smoker)[as.character(strat_data1$smoker)] /  
table(strat_data1$smoker)[as.character(strat_data1$smoker)]
```

#defining the survey design object

```
Strat1_design=svydesign(id=~1,strata = ~smoker,weights = ~weight,data  
=strat_data1 )
```

```
summary(Strat1_design)
```

#Estimating the mean values

```
svymean(~age,Strat1_design)
```

```
svymean(~bmi,Strat1_design)
```

```
svymean(~bloodpressure,Strat1_design)
```

```
svymean(~children,Strat1_design)
```

```
svymean(~claim,Strat1_design)
```

#Estimating the proportions

```
svymean(~gender,Strat1_design)
svymean(~diabetic,Strat1_design)
svymean(~smoker,Strat1_design)
svymean(~region,Strat1_design)
```

#Estimating the totals

```
svytotal(~gender,Strat1_design)
svytotal(~diabetic,Strat1_design)
svytotal(~smoker,Strat1_design)
svytotal(~region,Strat1_design)
```

#regression estimations

```
plot(strat_data1$claim,strat_data1$age)
plot(strat_data1$claim,strat_data1$bmi)
plot(strat_data1$claim,strat_data1$bloodpressure)
plot(strat_data1$claim,strat_data1$children)
SSLR1 = lm(claim~bloodpressure, data = strat_data1)
SSLR1
mean_claim=coef(SSLR1)[1] + coef(SSLR1)[2]*mean(data$bloodpressure)
mean_claim
```

#-----Sample2-----#

#draw the sample2

```
set.seed(12347)
strat_sample2=ssamp(df=data,n=n,strata=smoker,over=1)
strat_sample2
```

#obtaining weight

```
strat_data2$weight <- table(data$smoker)[as.character(strat_data2$smoker)] /  
table(strat_data2$smoker)[as.character(strat_data2$smoker)]
```

#defining the survey design object

```
Strat2_design=svydesign(id=~1,strata = ~smoker,weights = ~weight,data  
=strat_data2 )
```

```
summary(Strat2_design)
```

#Estimating the mean values

```
svymean(~age,Strat2_design)  
svymean(~bmi,Strat2_design)  
svymean(~bloodpressure,Strat2_design)  
svymean(~children,Strat2_design)  
svymean(~claim,Strat2_design)
```

#Estimating the proportions

```
svymean(~gender,Strat2_design)  
svymean(~diabetic,Strat2_design)  
svymean(~smoker,Strat2_design)  
svymean(~region,Strat2_design)
```

#Estimating the totals

```
svytotal(~gender,Strat2_design)  
svytotal(~diabetic,Strat2_design)  
svytotal(~smoker,Strat2_design)  
svytotal(~region,Strat2_design)
```

#regression estimations

```
SSLR2 = lm(claim~bloodpressure, data = strat_data2)
```

```
SSLR2
```

```
mean_claim=coef(SSLR2)[1] + coef(SSLR2)[2]*mean(data$bloodpressure)
```

```
mean_claim
```

##----sample1----##

#select Clusters

```
set.seed(12522)
```

```
selected_region1 <- sample(unique(data$region), size = 2, replace = FALSE)
```

```
selected_cluster_data1 <- data %>% filter(region %in% selected_region1)
```

#calculate sample size

```
n_cluster1<-ssampcalc(df=selected_cluster_data1, n=n, strata =region)
```

```
n_cluster1
```

#draw the sample1

```
cluster_sample1=ssamp(df=selected_cluster_data1,n=n,strata=region,over=1)
```

```
cluster_sample1
```

#obtaining weight

```
cluster_sample1$weight <- (nrow(selected_cluster_data1) /  
table(cluster_sample1$region)[as.character(cluster_sample1$region)])
```

```
#defining the survey design object
clust1_design=svydesign(id=~region,weights =
~weight,data=cluster_sample1)
summary(clust1_design)
```

#Estimating the mean values

```
svymean(~age,clust1_design)
svymean(~bmi,clust1_design)
svymean(~bloodpressure,clust1_design)
svymean(~children,clust1_design)
svymean(~claim,clust1_design)
```

#Estimating the proportions

```
svymean(~gender,clust1_design)
svymean(~diabetic,clust1_design)
svymean(~smoker,clust1_design)
svymean(~region,clust1_design)
```

#Estimating the totals

```
svytotal(~gender,clust1_design)
svytotal(~diabetic,clust1_design)
svytotal(~smoker,clust1_design)
svytotal(~region,clust1_design)
```

#regression estimations

```
plot(two_stage_sample1$claim,two_stage_sample1$age)
plot(two_stage_sample1$claim,two_stage_sample1$bloodpressure)
plot(two_stage_sample1$claim,two_stage_sample1$bmi)
```

```
plot(two_stage_sample1$claim,two_stage_sample1$children)
CSLR1 = lm(claim~bloodpressure,two_stage_sample1)
CSLR1
mean_claim=coef(CSLR1)[1] + coef(CSLR1)[2]*mean(data$bloodpressure)
mean_claim

#-----Sample2-----#
#select Clusters
set.seed(12541)
selected_region2 <- sample(unique(data$region), size = 2, replace = FALSE)
selected_cluster_data2 <- data %>% filter(region %in% selected_region2)

#calculate sample size
n_cluster2<-ssampcalc(df=selected_cluster_data2, n=n, strata =region)
n_cluster2

#draw the sample2
cluster_sample2=ssamp(df=selected_cluster_data2,n=n,strata=region,over=1)
cluster_sample2

#obtaining weight
cluster_sample2$weight <- (nrow(selected_cluster_data2) /
table(cluster_sample2$region)[as.character(cluster_sample2$region)])

#defining the survey design object
clust2_design=svydesign(id=~region,weights =
~weight,data=cluster_sample2)summary(clust2_design)
```

#Estimating the mean values

```
svymean(~age,clust2_design)
svymean(~bmi,clust2_design)
svymean(~bloodpressure,clust2_design)
svymean(~children,clust2_design)
svymean(~claim,clust2_design)
```

#Estimating the proportions

```
svymean(~gender,clust2_design)
svymean(~diabetic,clust2_design)
svymean(~smoker,clust2_design)
svymean(~region,clust2_design)
```

#Estimating the totals

```
svytotal(~gender,clust2_design)
svytotal(~diabetic,clust2_design)
svytotal(~smoker,clust2_design)
svytotal(~region,clust2_design)
```

#regression estimations

```
CSLR2 = lm(claim~bloodpressure,two_stage_sample2)
CSLR2
mean_claim=coef(CSLR2)[1] + coef(CSLR2)[2]*mean(data$bloodpressure)
mean_claim
```

#####-----Population-----#####

#Population size

N=nrow(data)

#mean values

mean(data\$age,na.rm=TRUE)

sqrt(var(data\$age,na.rm=TRUE)/N)

mean(data\$bmi,na.rm=TRUE)

sqrt(var(data\$bmi,na.rm=TRUE)/N)

mean(data\$bloodpressure,na.rm=TRUE)

sqrt(var(data\$bloodpressure,na.rm=TRUE)/N)

mean(data\$children,na.rm=TRUE)

sqrt(var(data\$children,na.rm=TRUE)/N)

mean(data\$claim,na.rm=TRUE)

sqrt(var(data\$claim,na.rm=TRUE)/N)

#proportions

data.frame((table(data\$gender))/length(data\$gender))

data.frame((table(data\$diabetic))/length(data\$diabetic))

data.frame(table(data\$smoker)/length(data\$smoker))

data.frame(table(data\$region)/length(data\$region))

#totals

table(data\$gender)

table(data\$diabetic)

table(data\$smoker)

table(data\$region)

###-----graphical analysis-----###

#Bar Charts

```
byg<-svyby(~claim, by=~gender,Strat1_design, svymean)
```

```
barplot(byg, names.arg=c("Female","Male"),col="light blue",ylab="mean  
claim",xlab="gender type",main="Mean Insurance Claim by Gender")
```

```
byr<-svyby(~claim, by=~region,Strat1_design, svymean)
```

```
barplot(byr,names.arg = c("Northeast", "Northwest", "Southeast",  
"Southwest"),col="light blue",ylab="mean claim",xlab="region type",main="Mean  
Insurance Claim by Region")
```

```
bysm<-svyby(~claim, by=~smoker,Strat1_design, svymean)
```

```
barplot(bysm, names.arg=c("non_smoker","smoker"),col="light blue",ylab="mean  
claim",xlab="smoking status",main="Mean Insurance Claim by Smoking")
```

```
byd<-svyby(~claim, by=~diabetic,Strat1_design, svymean)
```

```
barplot(byd,names.arg=c("non_diabetic","diabetic"),col="light blue",ylab="mean  
claim",xlab="Diabetic status",main="Mean Insurance Claim by diabetic")
```

#Histograms

```
svyhist(~claim,SRS_Sample1_design,freq =TRUE,col="light green")
```

```
svyhist(~bloodpressure,SRS_Sample1_design,freq =TRUE,col="light blue")
```

```
svyhist(~bmi,SRS_Sample1_design,freq =TRUE,col="light yellow")
```

```
svyhist(~claim,Strat1_design,freq =TRUE,col="light green")
```

```
svyhist(~bloodpressure,Strat1_design,freq =TRUE,col="light blue")
```

```
svyhist(~bmi,Strat1_design,freq =TRUE,col="light yellow")
```

```
svyhist(~claim,clust1_design,freq =TRUE,col="light green")
svyhist(~bloodpressure,clust1_design,freq =TRUE,col="light blue")
svyhist(~bmi,clust1_design,freq =TRUE,col="light yellow")
```

```
hist(data$claim, freq = TRUE, col = "light green")
hist(data$bloodpressure, freq = TRUE, col = "light blue")
hist(data$bmi, freq = TRUE, col = "light yellow")
```

#Box plots

```
svyboxplot(claim~gender,SRS_Sample1_design,all.outliers=TRUE,col="light
yellow",xlab="Gender type",ylab="claim amount",main="claim vs Gender")
svyboxplot(claim~smoker,SRS_Sample1_design,all.outliers=TRUE,col="light
yellow",xlab="smoking status",ylab="claim amount",main="claim vs smoking")
svyboxplot(claim~region,SRS_Sample1_design,all.outliers=TRUE,col="light
yellow",xlab="Region ",ylab="claim amount",main="claim vs region")
svyboxplot(claim~diabetic,SRS_Sample1_design,all.outliers=TRUE,col="light
yellow",xlab="Diabetic status",ylab="claim amount",main="claim vs diabetic")
```

#Scatter plots

```
svyplot(claim~age,design=SRS_Sample1_design,
style="bubble",xlab="age",ylab="claim")
svyplot(claim~bloodpressure,design=SRS_Sample1_design,
style="bubble",xlab="bloodpressure",ylab="claim")
svyplot(claim~bmi,design=SRS_Sample1_design,
style="bubble",xlab="BMI",ylab="claim")
svyplot(claim~children,design=SRS_Sample1_design,
style="bubble",xlab="Number od children",ylab="claim")
```

